Multi-Epoch Photometry & Spectroscopy of Luminous Blue Variables (LBVs) in M31 & M33

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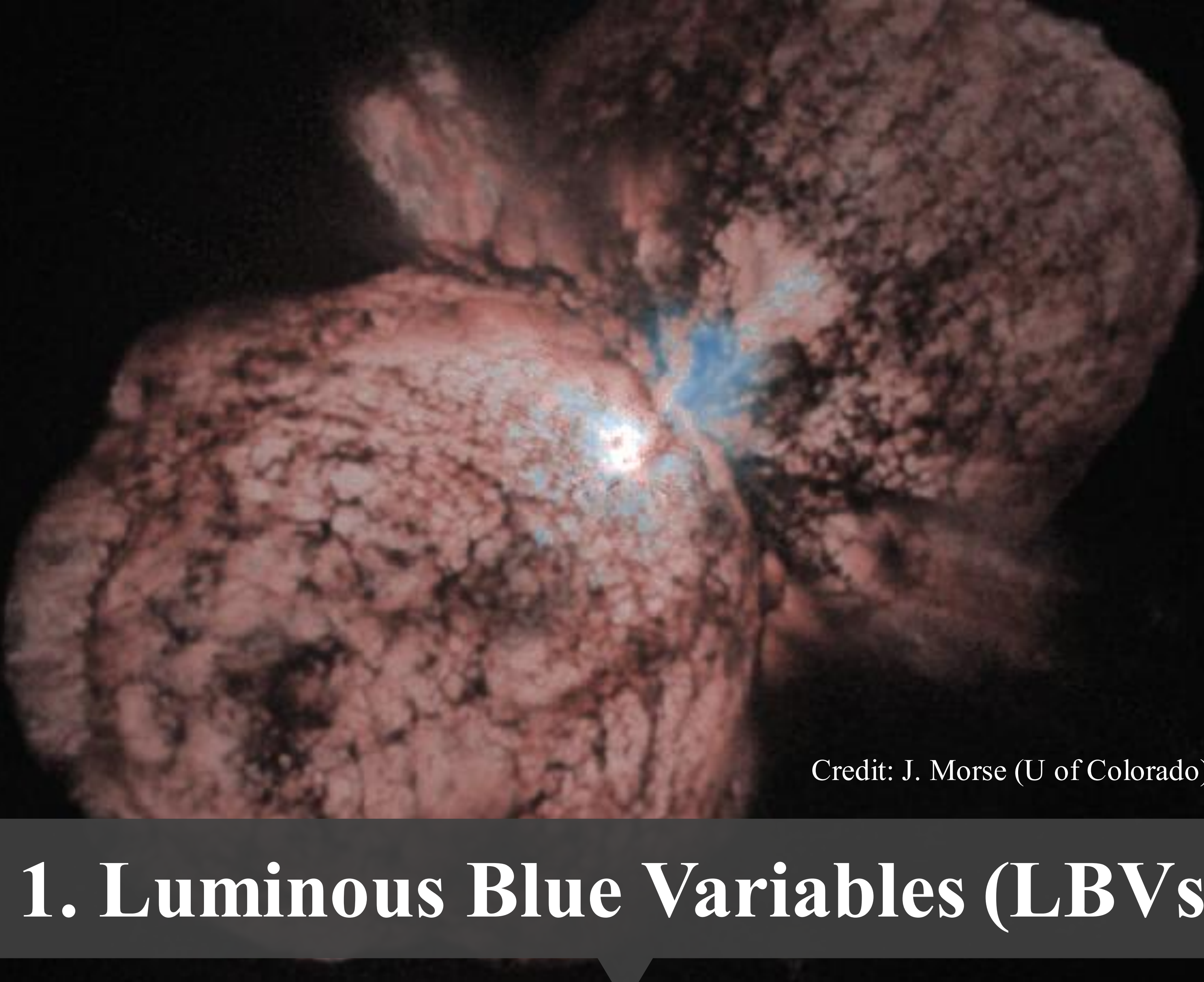
Agenda

01	Luminous Blue Variables
02	Methodology
03	Preliminary Results
04	Future Implications



Credit: J. Morse (U of Colorado) & Hubble

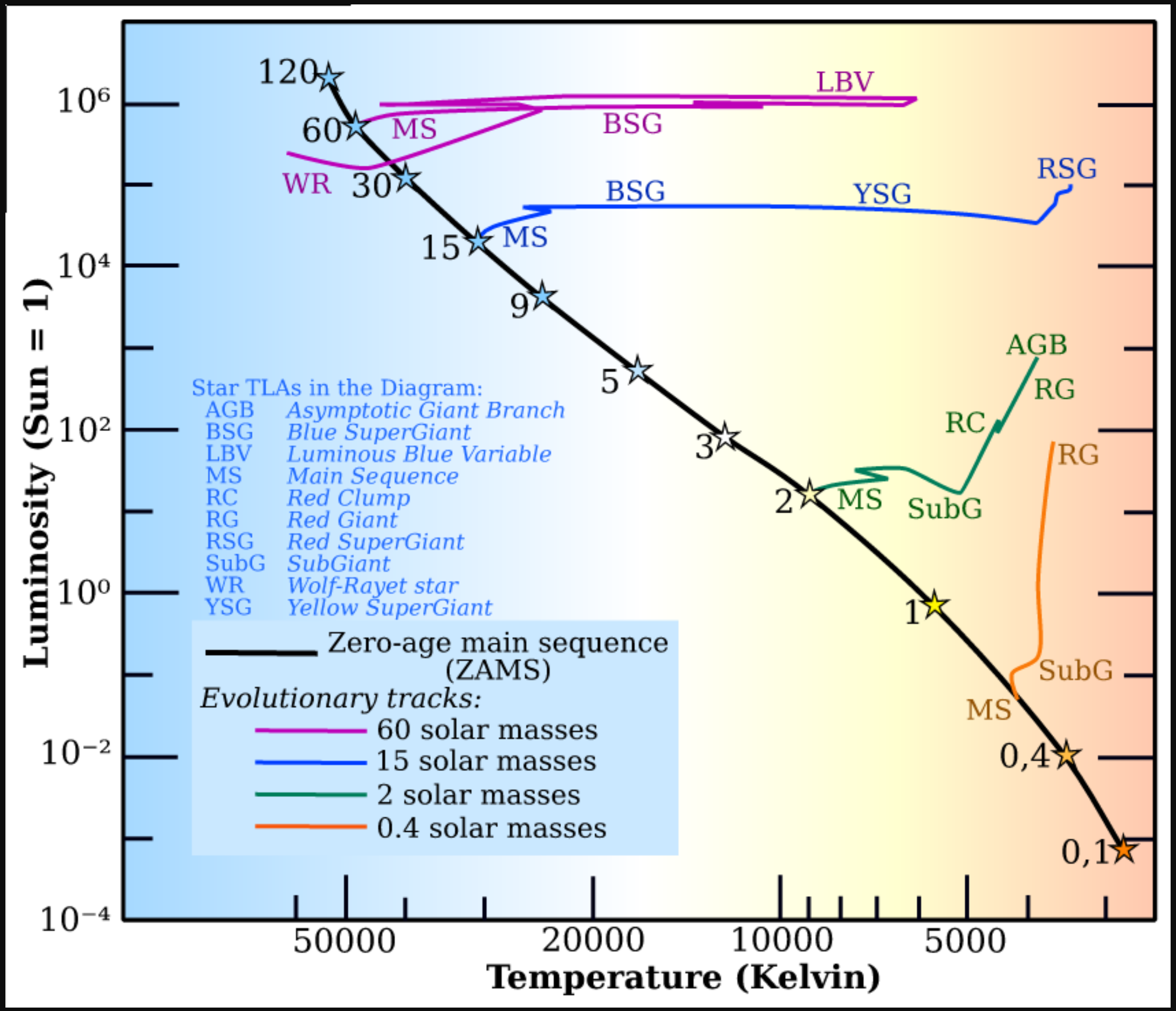
1. Luminous Blue Variables (LBVs)

Post-MS Evolution of Massive Stars

- Post-MS evolution for stars with $M_{ZAMS} \geq 30\ M_{\odot}$ is different from those with lower mass
  - Evolve with constant bolometric luminosity
  - Vary depending on luminosity

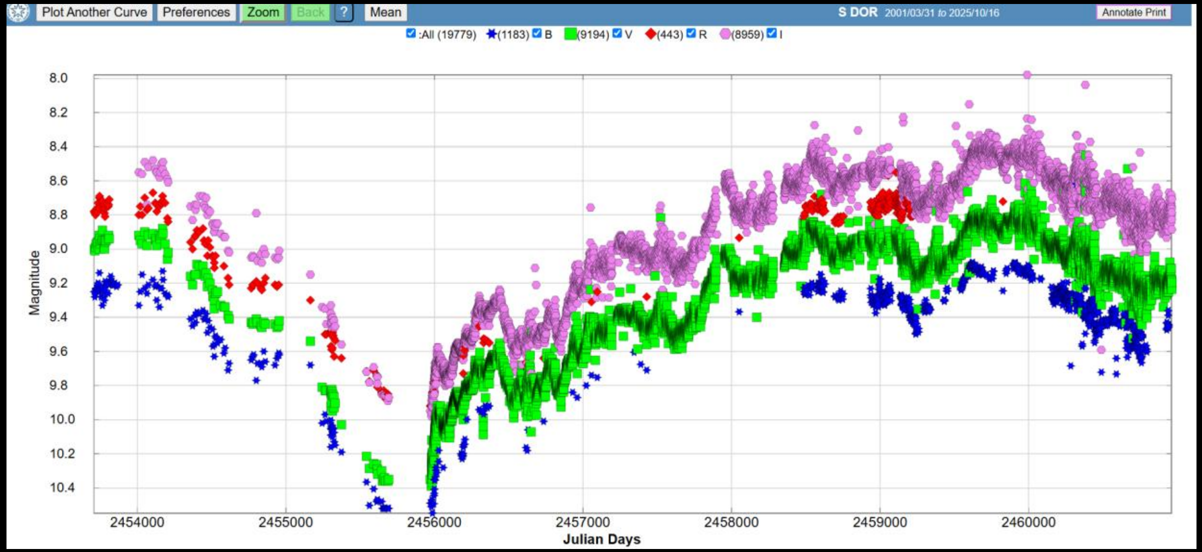
    E.g., stars with $\log\left(\frac{L}{L_{\odot}}\right) > 5.8$ may not become RSGs (Humphreys & Davidson 1994)
  - Mass loss also plays a *deterministic* role (Smith & Tombleson 2015)
- One of the typical evolutionary scheme is: (Sholukhova et. al. 2018, see also Smith & Tombleson 2015)

  O → Of/ WNH → **<u>LBV</u>** → WN → WC → SN



LBVs

- Also classified as Hubble-Sandage variables (Hubble & Sandage 1953) or S Dor variables (Wolf 1989)
- Have the **highest** mass loss rates of any stars (Smith et. al. 2020)
- Unstable, massive, evolved stars with irregular outbursts, leading to the formation of a "pseudo-photosphere" (Szeifert et. al. 1996)

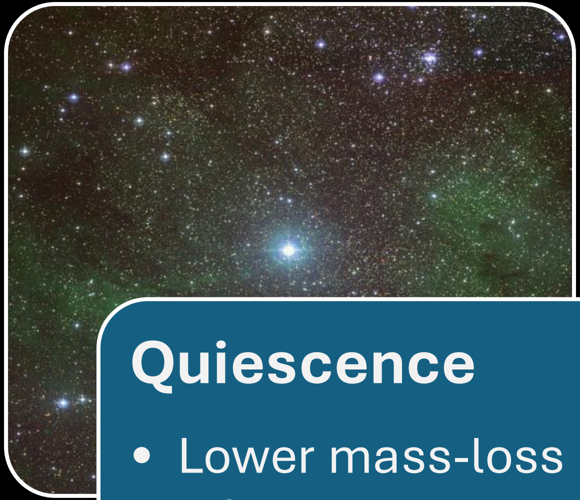


(Credit: AAVSO enhanced LCG)

Pseudo-photosphere?

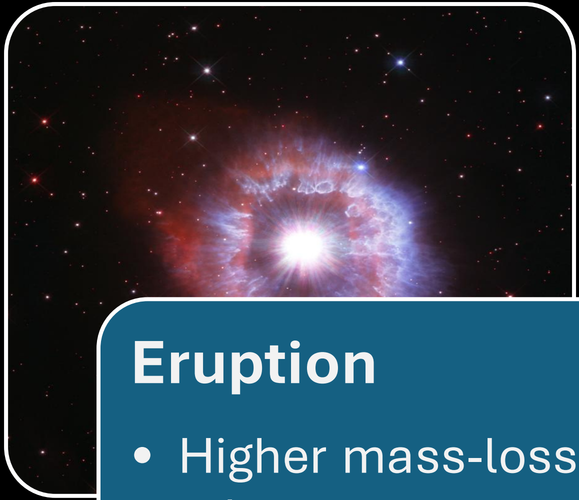
(Credit: NASA, ESA, STScI)

Pseudo-photosphere?



Quiescence

- Lower mass-loss rate
- “Normal” high temperature (> 15,000 K)



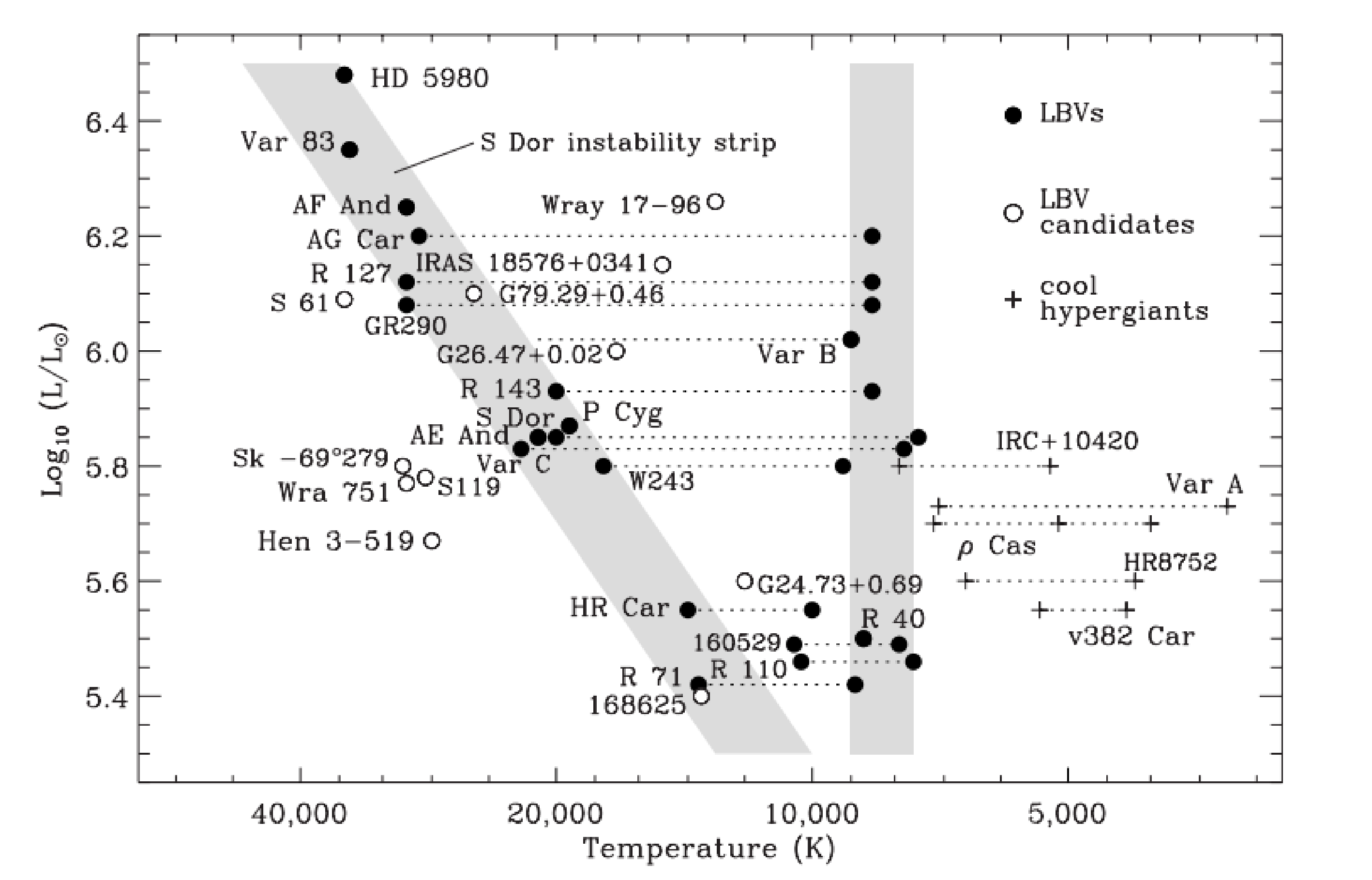
Eruption

- Higher mass-loss rate
- Cool (7000-9000 K) and dense ($N \sim 10^{11}$ cm^{-3}) expanding envelope

→ Brightening in visible wavelengths at **constant bolometric luminosity**

(Szeifert et. al. 1996, see also Humphreys & Davidson 1994)

S Dor instability & constant-temperature strips



(Smith, Vink, & de Koter 2004)

2. Methodology

Methodology

- Photometry Data Acquisition:
  - All LBV and LBV candidates: John Martin at the University of Illinois Springfield Henry R. Barber Research Observatory
  - For M31 AE And & AF And and M33 Var 2 and Var C: American Association for Variable Star Observers
  - Also obtained data for M31 AE And, AF And, Var 15, Var A-1 and M33 Var B, Var C, Var 83, Var 2 from Szeifert et al. (1996)
- Spectroscopy from the MMT Observatory
- Data Analysis:
  - Calculate color index
  - Apply color correction: adopting A_V from Humphreys et. al. (2017)
  - Convert to absolute V-band magnitude
  - Plot a Color-Magnitude Diagram ($B - V$ vs. M_V)
  - Plot the change in $B - V$ against the change in M_V

(Credit: Steward Observatory)

Methodology

Table 2
Extinction, Luminosities, and Temperatures—LBVs and Candidates

Star	A_v (stars) (mag)	A_v (H I) (mag)	Adopted A_v (mag)	M_v (mag)	T (K)	M_{Bol} (mag)	Stellar Group[a]
			M31				
LBVs							
AE And	0.9(1)	0.9	0.9	−7.9:	20000:	−9.4:	A170, H II
AF And	…	1.1	1.1	−8.2:	28000:	−10.7:	…
Var A-1[b]	1.4(1)	1.9	1.4	−8.7	21700	−10.6:	A42
Var 15[b]	…	1.3	1.3	−7.3:	20200:	−9.1:	A38
M31-004526.62[c]	1.6(1)	1.3	1.5	…	…	−9.65	A45, H II
LBV Candidates							
M31-003910.85	1.0(3)	1.0	1.0	−7.2	17400	−8.7	A127
M31-004051.59[d]	1.7	1.6	0.6	−8.1	9000	−8.4	A82
M31-004411.35	2.3(1):	0.8	2.3:	−8.6	17400	−10	A10
M31-004425.18[e]	…	0.7	0.7		18000	−8	isolated(A9)
M31-004444.00	0.5(3)	1.0	0.5	−5.9	20700	−7.7	A54
			M33				
LBVs							
Var B	0.4	0.3	0.4	−7.7	24000:	−10.0	A142
Var C[f]	0.5(2)	0.4	0.5	−8.6	20000:	−9.8	H II
Var 83	0.8(2)	0.9	0.8	−8.8:	28000:	−11.1:	A101,A103, H II
Var 2	0.9(1)	1.0	0.9	−7.2	28000:	−10.0:	A100
LBV Candidates							
M33C-2976	0.4(2)	0.5	0.4	−5.9	17000	−7.2	A124
M33C-4174	0.8(3)	0.9	0.8	−7.3	19900	−9.0	A130
UIT008[g]	0.4(1)	0.8	0.6	−7.5	34000	−10.7	H II, A27
M33C-14239	0.3(2)	0.4	0.4	−7.6	16600	−9.0	Spiral arm
M33C-4640[h]	0.6(2)	0.6	0.6	−8.1	9000	−8.3	A128
M33C-25255	0.7(3)	0.6	0.6	−6.3	24500	−8.6	A137
M33-013317.22	0.7(2)	0.5	0.7	−6.5	29100	−10.0	A17
M31-013334.06	…	0.6	0.6	−7.6	19800	−9.3	…
M33C-7024	0.3(3)	0.3	0.3	−6.3	25200	−8.6	Spiral Arm
M33C-15235	1.1(4)	0.8	1.0	−7.8	29200	−10.4	A64
M33C-5916	0.7 (1)	0.6	0.7	−6.9	34000	−10.0	A6
M33-013406.63[i]	0.8(1)	0.7	0.7:	−9.1:	29200:	−12.6:	H II
M33C-21386	0.9(4)	1.0	0.9	−8.2	…	…	A71
M33C-10788	0.8(3)	0.5	0.8	−7.3	32000	−10.3	A100
M33-013424.78	…	0.9	0.9	−8.8	13700	−9.4	A102
M33C-20109	2.4(1):	…	2.4:	−8.5	…	…	…
M33-013432.73	0.6(2)	0.6	0.6	−6.0	21700	−9.0	A84, N604
M33C-16364	0.4(6)	0.5	0.4	−6.7	31100	−9.4	A88
V532/GR290[j]	…	0.6	0.6	var	42000−22000	−10.4–10.7	A89:

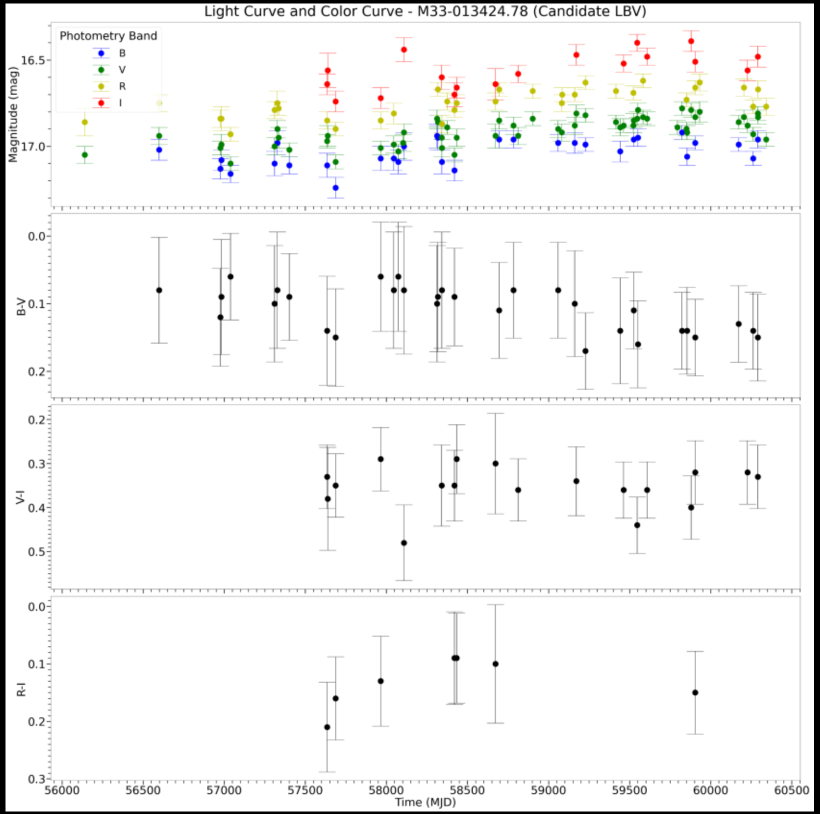
Methodology

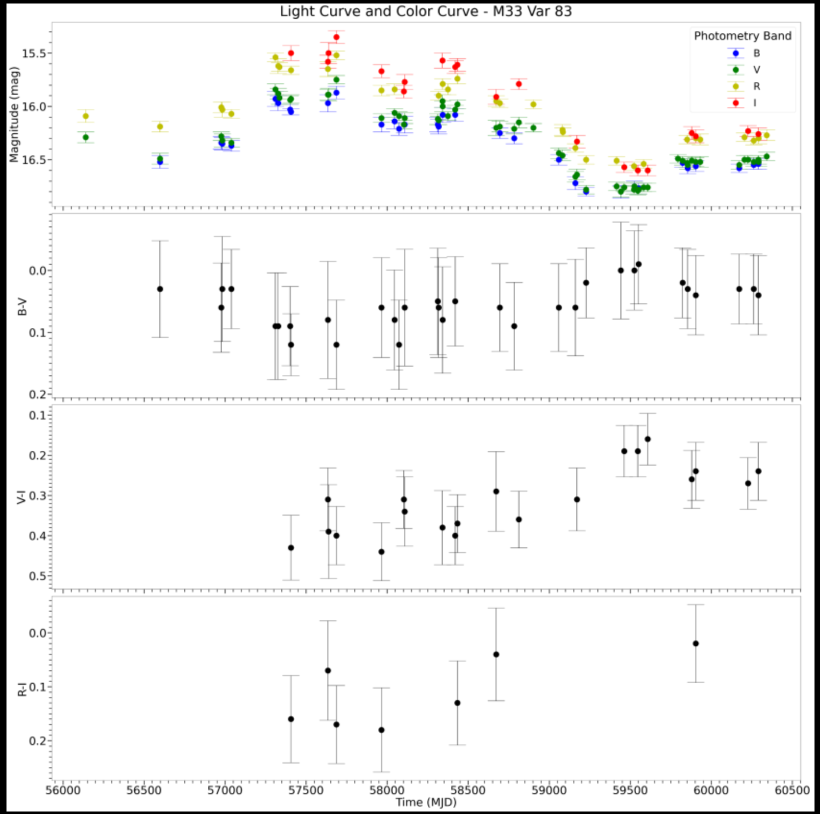
Table 3
Extinction, Luminosities, and Temperatures—B[e] Supergiants and Warm Hypergiants

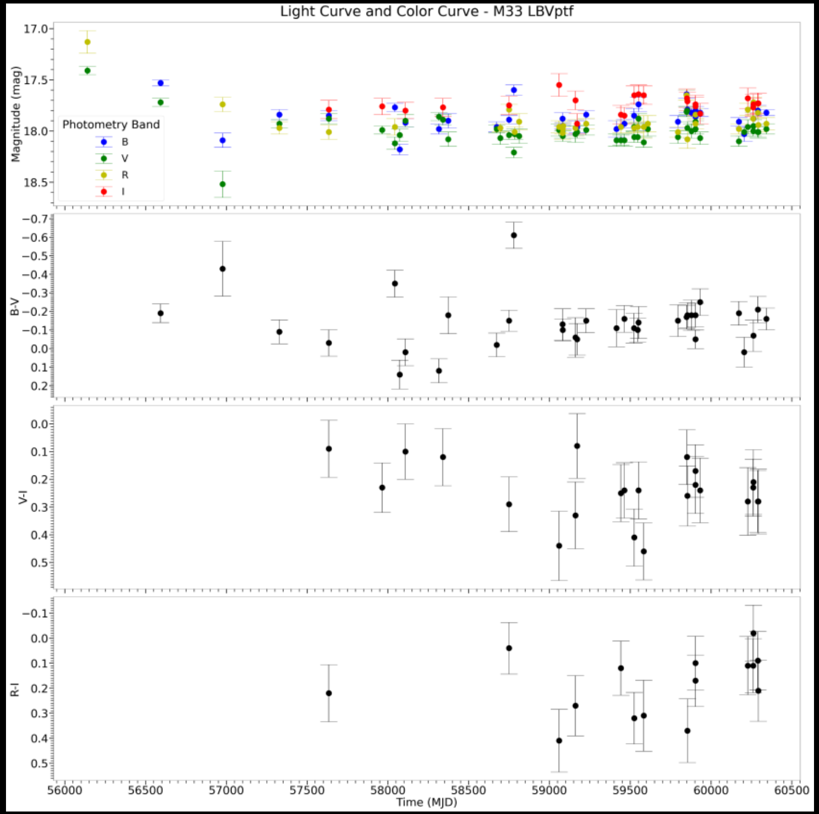
Star	A_v (stars) (mag)	A_v (H I) (mag)	Adopted A_v (mag)	M_v (mag)	T (K)	M_{Bol} (mag)	Stellar Group[a]
			M31				
B[e] Supergiants							
M31-004043.10[b]	1.5(1)	0.8	1.0	−6.8	16500	−8.4	A120
M31-004057.03	···	0.7	0.7	−6.3	7700:	−6.2:	···
M31-004220.31[b]	···	1.1	1.1	−6.6	12500	−7.7	···
M31-004221.78[b]	···	0.95	1.0	−5.8	7200:	−7.0	···
M31-004229.87[b]	1.4(2)	2.1	1.4	−7.0	17600	−8.8	···
M31-004320.97[b]	···	1.5	1.5	−6.7	10500	−7.9	···
M31-004415.00[b]	···	0.9	0.9	−7.0	17300	−8.9	···
M31-004417.10[b]	0.7(1)	1.7	0.7	−8.0	13800	−9.0	A32:
M31-004442.28[b]	1.2(1)	1.7	1.2	−5.9	26900	−8.6	···
Warm Hypergiants							
M31-004322.50[b]	1.5(1)	1.1	1.5	−5.6	8200	−6.2	Spiral Arm
M31-004444.52[b]	1.5(3)	1.3	1.5	−7.8	6600	−8.4	A41
M31-004522.58[b]	1.2(1)	1.1	1.2	−7.1	11000	−7.7	A46
M31-004621.08[b]	1.2(1)	0.3	1.2	−7.6	10000	−8.0	A104:
			M33				
B[e] Supergiants							
M33-013242.26	0.7(2)	0.8	0.7	−7.8	22900:	−9.3	A118
M33-013324.62[b]	···	0.4	0.4	−5.3	22800	−7.6	···
M33C-7256[b]	0.5(3)	0.3	0.5	−5.6	18400	−8.0	A14
M33C-6448	0.5(2)	0.8	0.5	−6.9	17300	−8.3	A9
M33C-24812[b]	···	0.7	0.7	−6.2	20000	−8.0	A35
M33C-15731[b]	1.2(2)	0.7	1.2	−8.9	21700	−11.0:	A64(bulge)
M33-013426.11[b]	···	0.6	0.6	−6.1	12300	−7.2	···
M33-013459.47[b]	···	0.7	0.7	−6.8	23200	−9.0	···
M33-013500.30[b]	0.3(3)	0.7	0.3	−5.5	16200	−7.2	A88
Warm Hypergiants							
Var A[b,c]	0.6–0.9	1.0	···	···	7000:	−9.5	A130
B324[d]	0.9(3)	0.8	0.8	−10.2	8000	−10.1	A67
N093351[b,e]	0.45(2)	0.8	0.45	−8.8	7800	−8.8	A142
N125093[b,e]	0.9(1)	0.8	1.6	−8.8	7500	−8.9	A105
J013358.05+304539.9[b,f]	0.8(2)	0.6	0.8	−9.2	5000–6000	−9.3:	A68

3. Preliminary Results

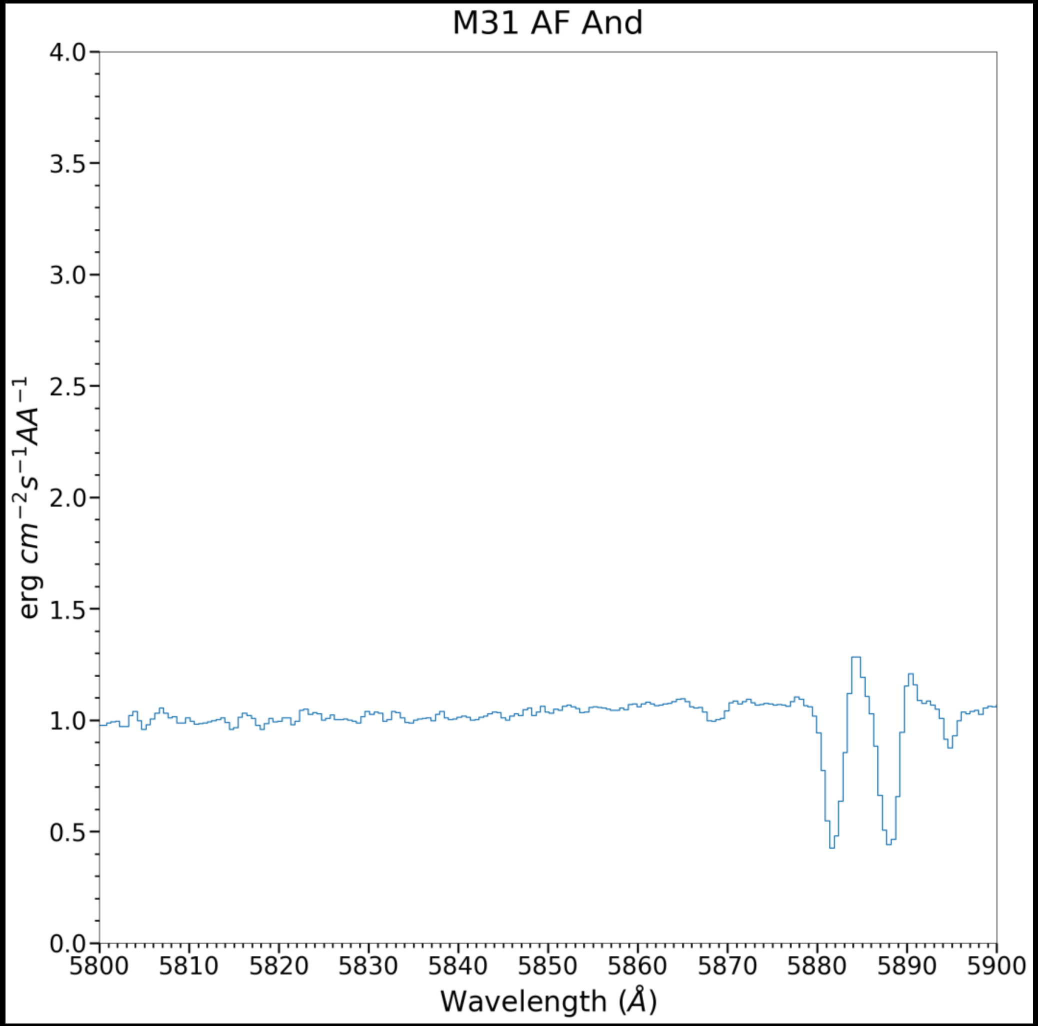
Light Curves and Color Curves

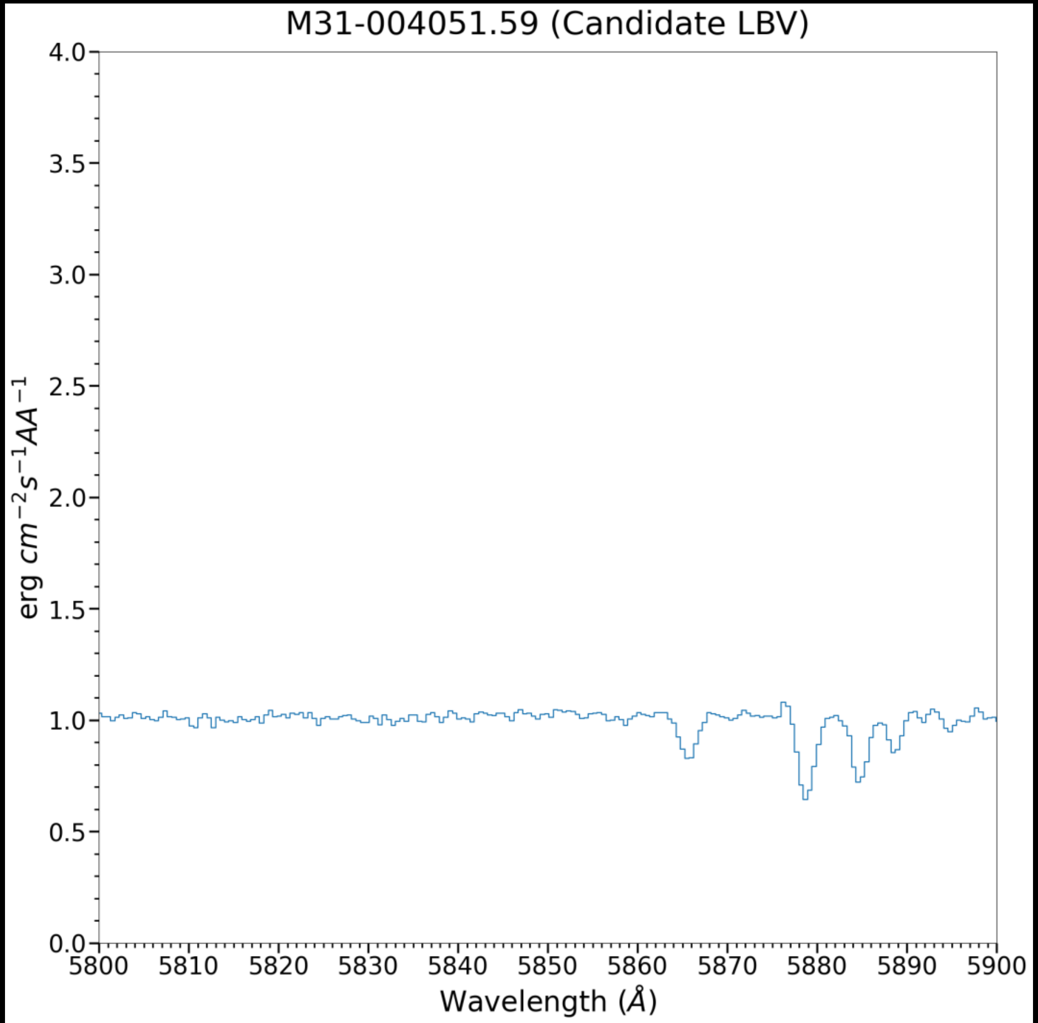




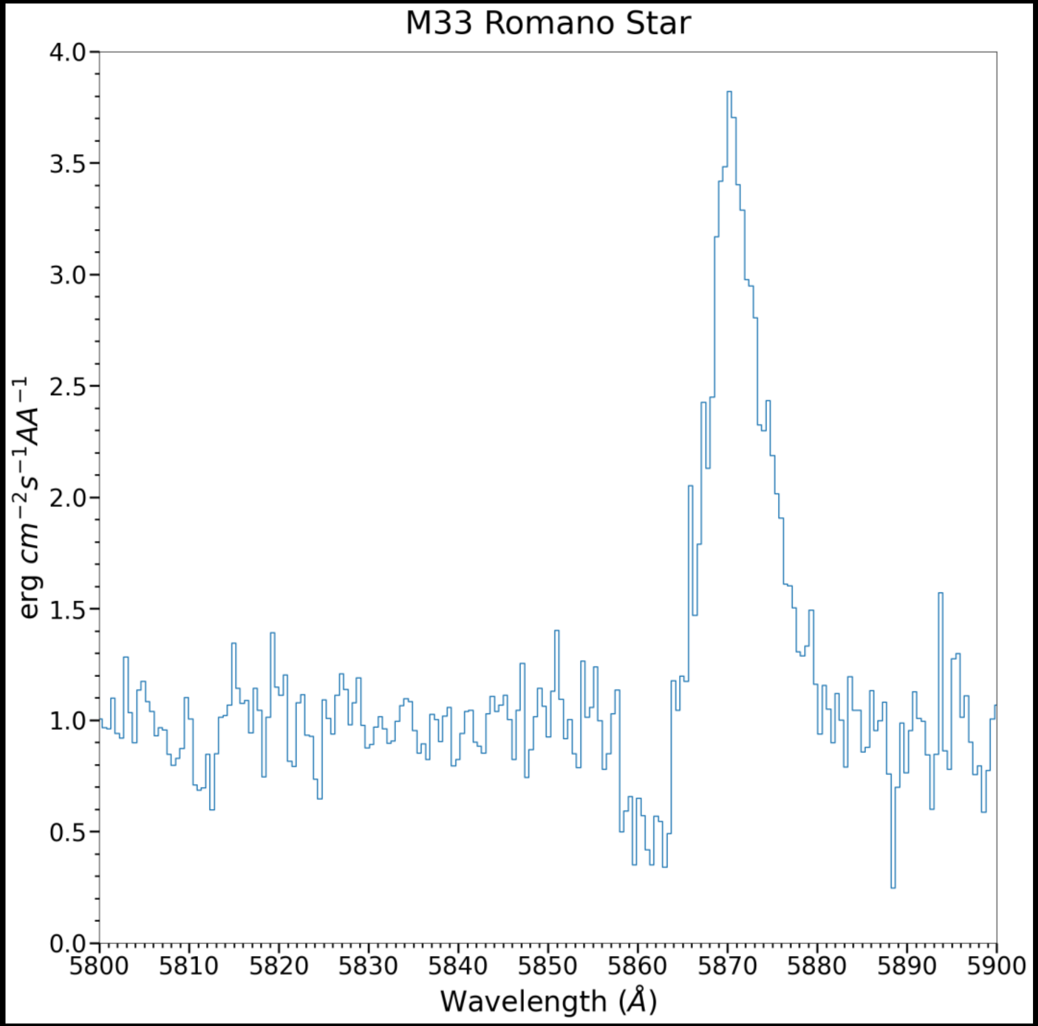


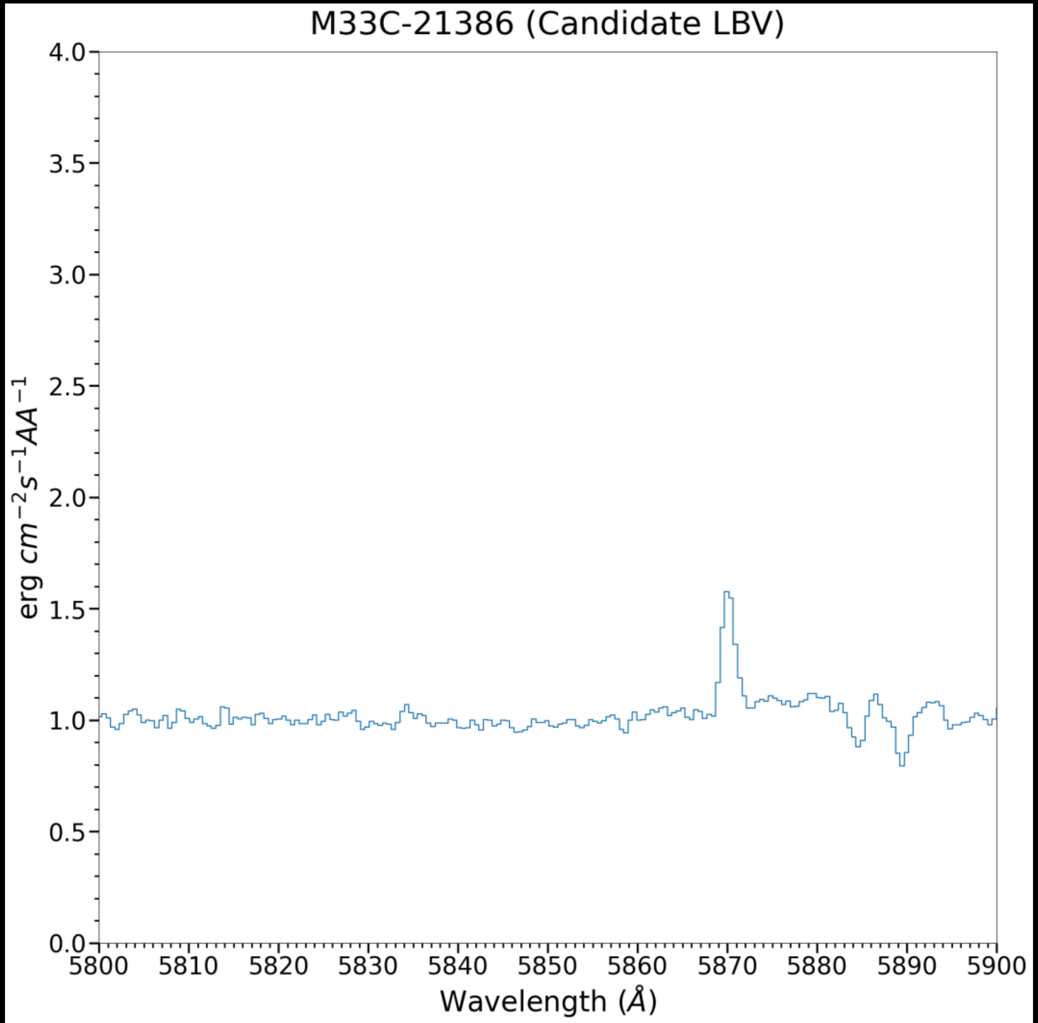
M31 Objects' Spectra



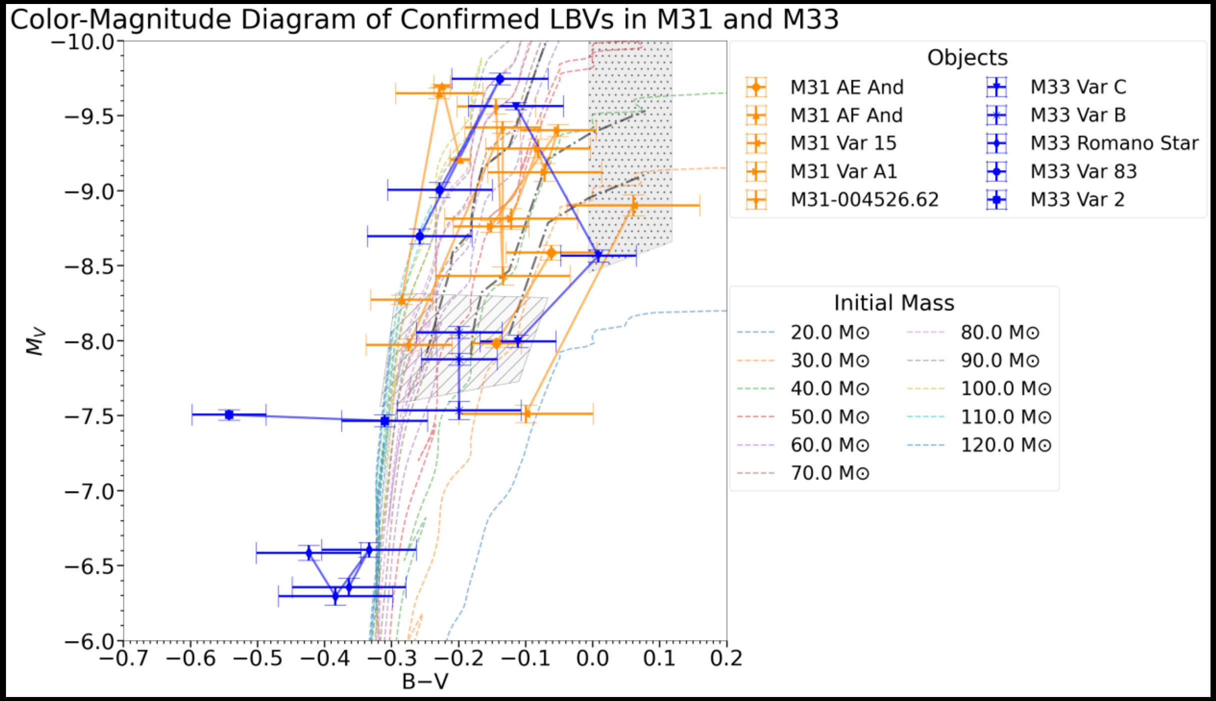


M33 Objects' Spectra

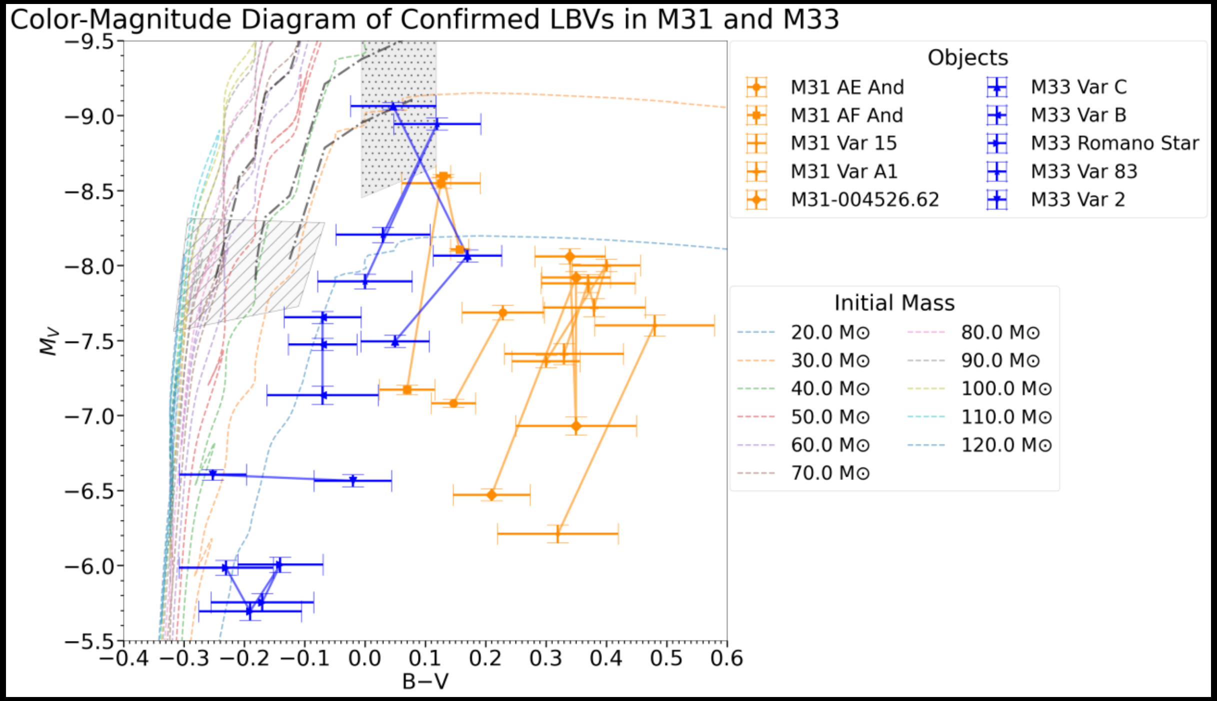




Color-Magnitude Diagram

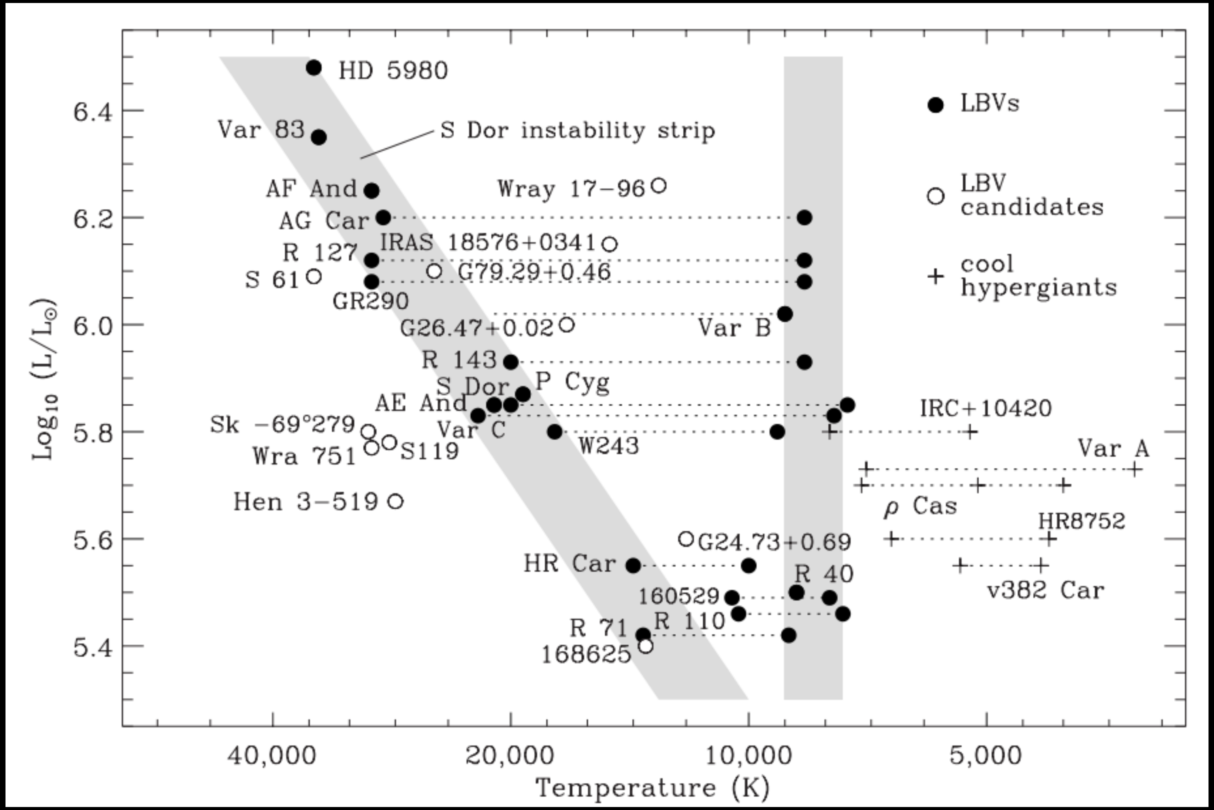


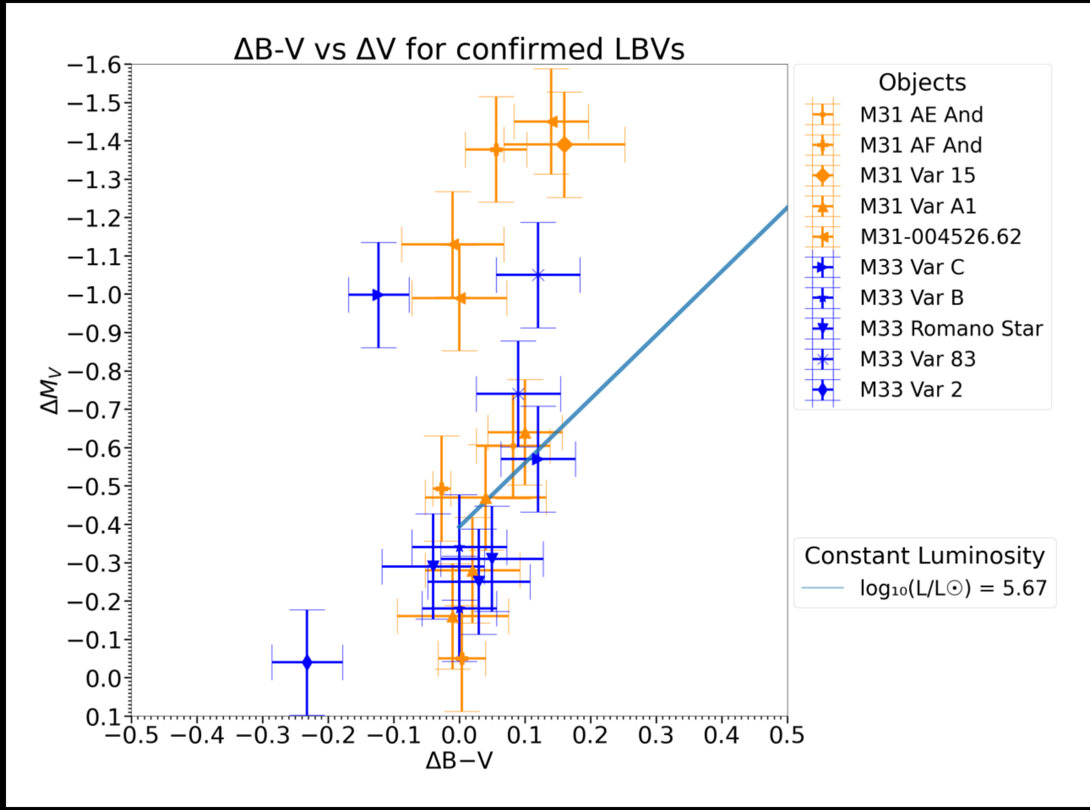
Adopted A_V from Humphreys (2017)



No color-correction

Δ(B – V) vs. ΔV

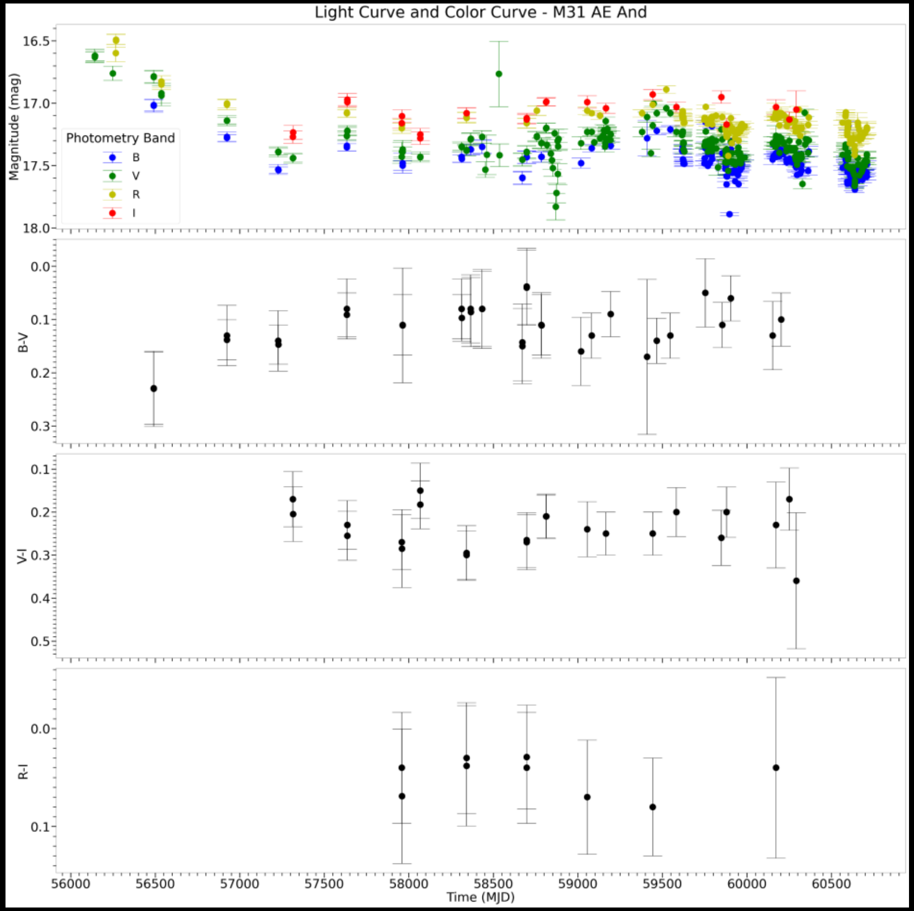

HD 5980
Var 83
S Dor instability strip
AF And
AG Car
Wray 17−96
IRAS 18576+0341
R 127
S 61
G79.29+0.46
GR290
G26.47+0.02
Var B
R 143
S Dor
P Cyg
AE And
Var C
Sk −69°279
Wra 751
S119
W243
IRC+10420
Var A
Hen 3−519
ρ Cas
HR8752
G24.73+0.69
HR Car
R 40
160529
R 110
R 71
168625
v382 Car
LBVs
LBV candidates
cool hypergiants
Log₁₀ (L/L☉)
Temperature (K)

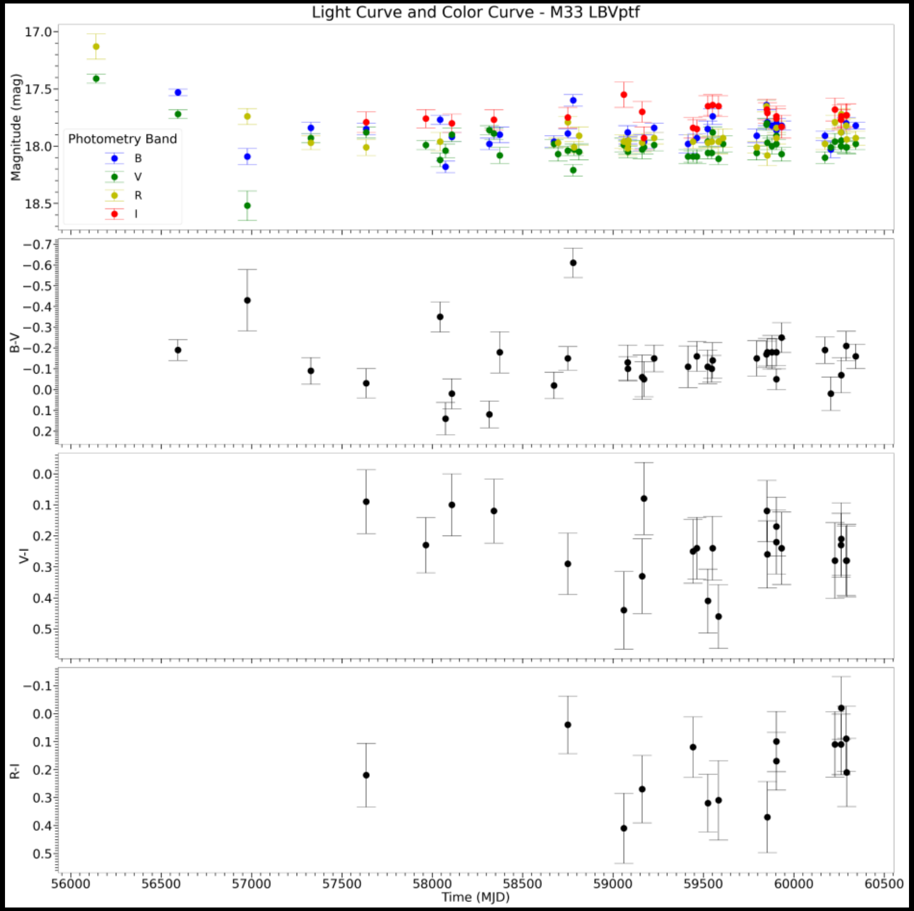

ΔB-V vs ΔV for confirmed LBVs
Objects
M31 AE And
M31 AF And
M31 Var 15
M31 Var A1
M31-004526.62
M33 Var C
M33 Var B
M33 Romano Star
M33 Var 83
M33 Var 2
Constant Luminosity
log₁₀(L/L☉) = 5.67
ΔM_V
ΔB−V

4. Future Implications

Future Implications

- Obtain more photometry to capture missing outbursts



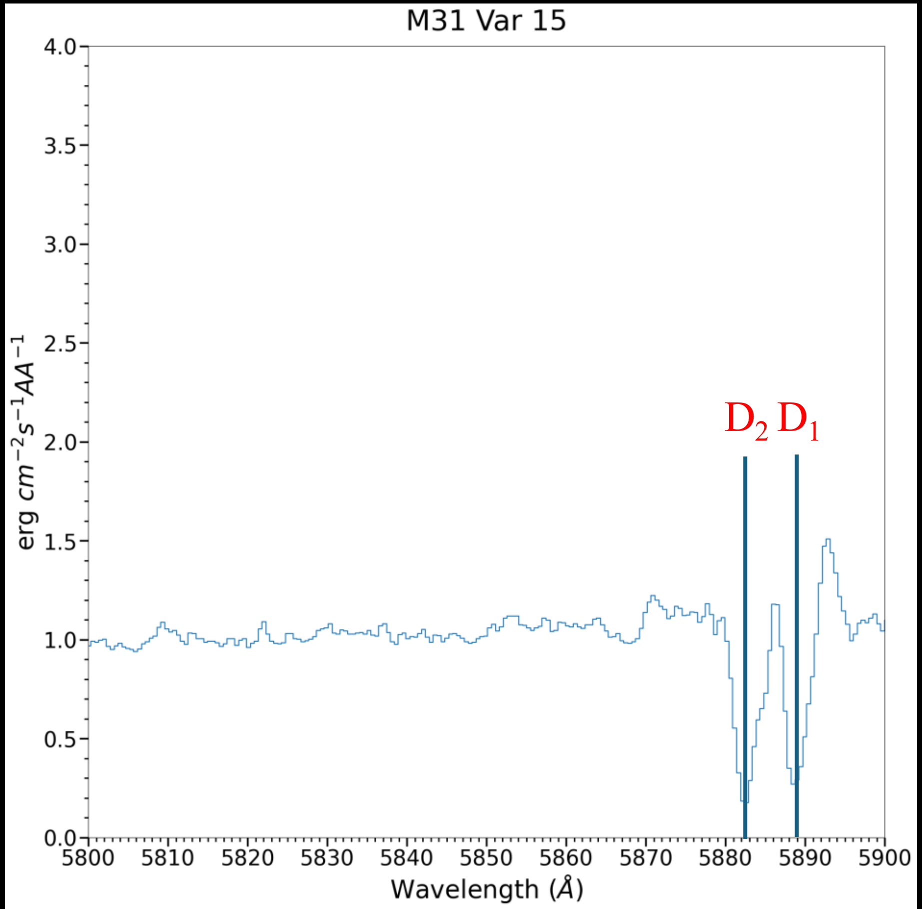


Future Implications

- A different color correction method with Na I D absorption lines

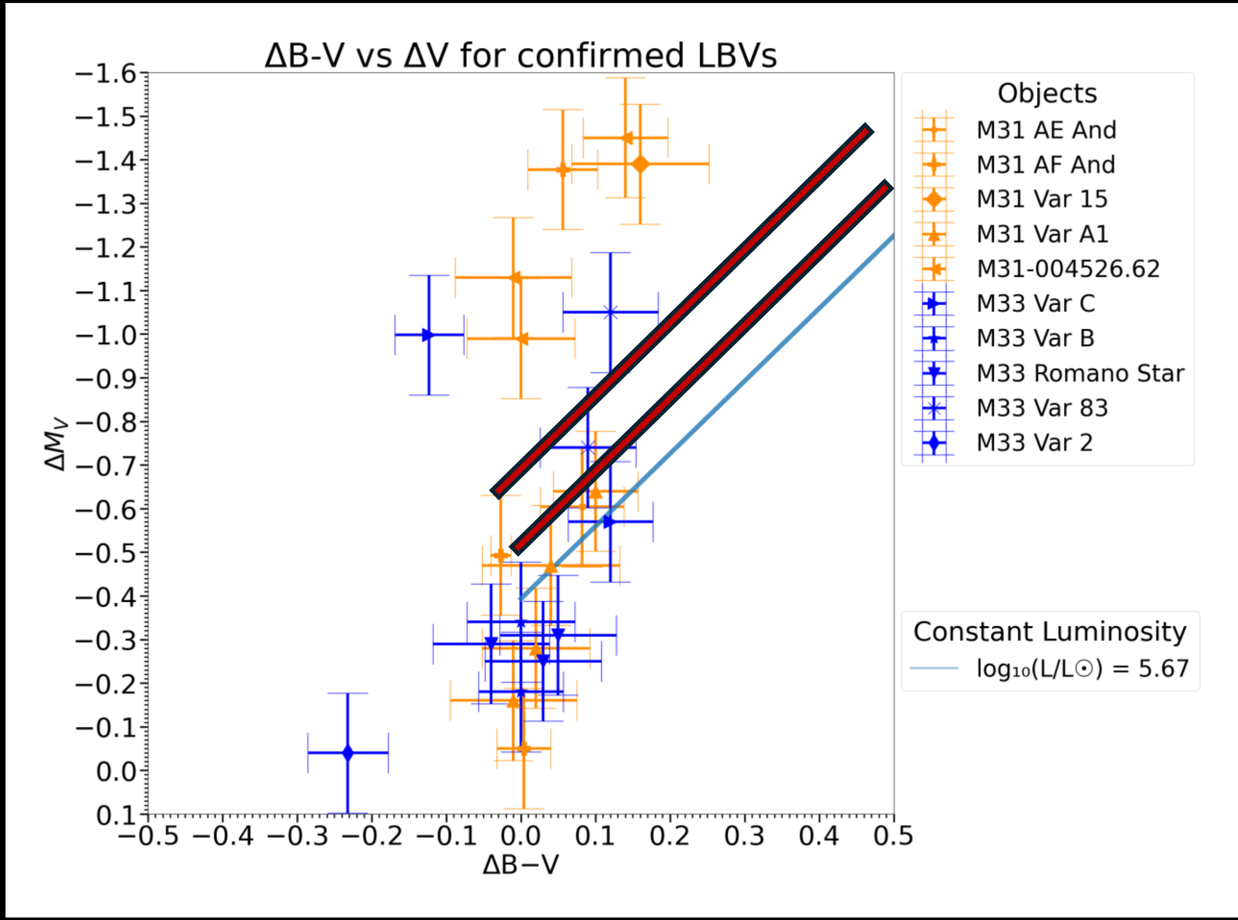
$$\log_{10}(E_{B-V}) = 1.17 \times EW(D_1 + D_2) - 1.85 \pm 0.08$$

(Podnanski et al. 2012)



Future Implications

- Do they get brighter with constant bolometric luminosity?



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Takeaways

- LBVs are evolved, massive stars that brightens in optical wavelengths <u>at constant bolometric luminosity</u>
- Current photometry shows that they become reddened as they become brighter during eruptions
- Future implications include obtaining more photometry and applying a new method of color-correction using sodium absorption lines